Modelling the Impact of Climate Change on Soil Moisture and UK's Future Agriculture

Project 2
27 February 2025

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Agenda

1. **Project Background**
2. Project Scope
3. Methodology
4. The Datasets
5. Data Preprocessing
6. Exploratory Data Analysis
7. Model Development
8. Key Findings
9. Further Research

Project Background

This project focuses on understanding near-surface air temperature trends in the UK using ensemble climate model simulations from 2006 to 2080. The analysis is based on six NetCDF datasets generated by the same climate model under slightly varied conditions (ensembles).

IPCC states that global temperatures could increase by up to 4.8°C by the end of the century, substantially raising the risk of severe climate-related impacts.

This study leverages 6 datasets containing 3 dimensional data across the UK, including variables such as TREFHT amongst several others.

ERA5 data on topsoil temperature and moisture was also used across the same coordinates and merged to analyse the impact on agriculture.

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Project Scope

(1) Analyse Trend & Patterns of TREFHT

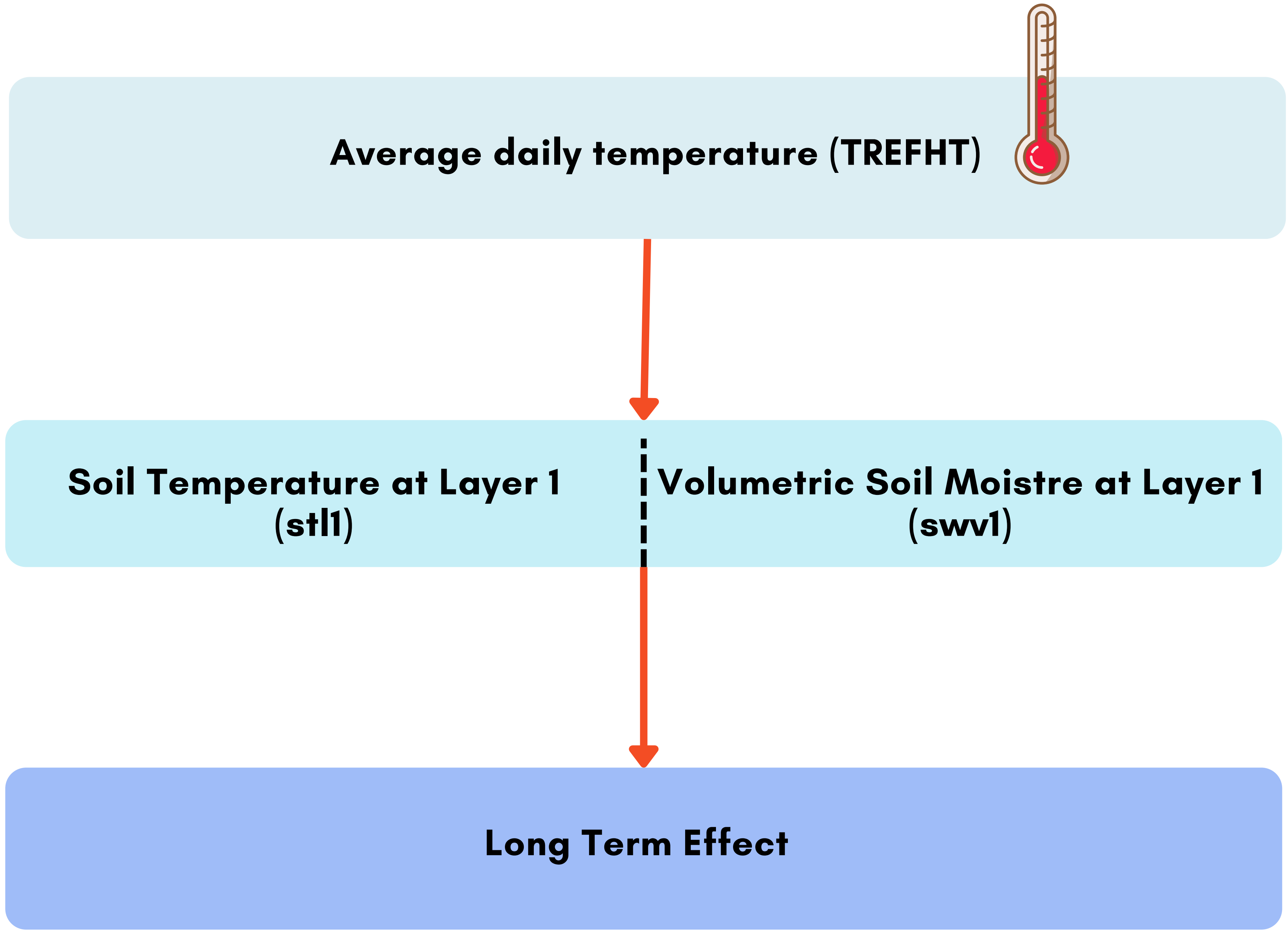
Combine the six given datasets and analyse seasonal, long-term and regional trends of TREFHT over time. TREFHT was chose over TREFMAX_U due to its cover of rural regions, and therefore relevance to agriculture

(2) Analyse Trend & Patterns of Topsoil

Analyse soil temperature at layer 1 (stl1) and volumetric soil water at layer 1 (swvl1) due to their strong influence on biological processes such as seed germination, availability of nutrients, root development and several others.

(3) What Is The Long-Term Effect?

Build a model to analyse the long-term effect of temperature on topsoil and its potential impacts on the agriculture industry.



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Project Methodology and Delivery

The project follows a 6-phase methodology comprehensively evaluating datasets to build accurate predictive model

Project Phase	Activity Overview	Deliverable
Combine Provided Datasets	Take the average of each variable value across 6 datasets to reduce variability and enhance robustnes	Merged dataset
Data Preprocessing	Check data type, missing values, duplicates, variable units, incorrect values	Preprocessed merged dataset
Gather External Dataset + Preprocessing + Merge	Gather ERA5 data on soil metrics, preprocess, align timestamps and coordinates before merging.	Enriched dataset
Exploratory Data Analysis	Examine pattern in target variable (TREFHT), how it correlates with other variables, stl1 and swvl1.	Trends and key factors affecting target variable
Build Predictive Models	Divide the dataset into training period (2006-2014) and testing period (2015-2080). Two models will be explored: XGBoost and Prophet.	Best predictive model with the highest accuracy
Draw Conclusion	The results will be used to analyse the effect of rising temperatures on agriculture. How will farming conditions look in 2080?	Key findings and future recommendations

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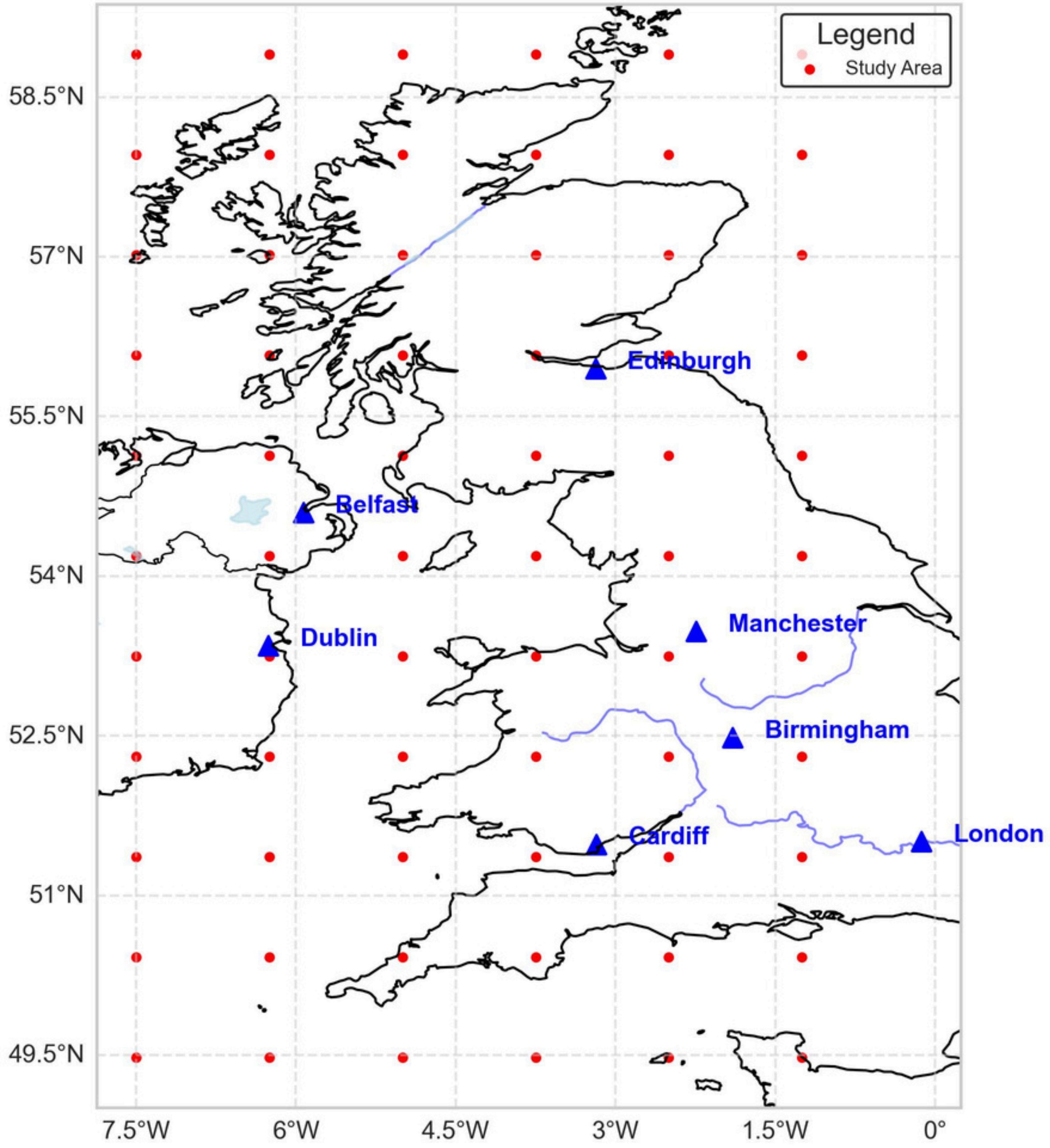
The Provided Datasets:
Climate Data from Year 2006 - 2080

- 003_2006_2080_352_360.nc
- 004_2006_2080_352_360.nc
- 005_2006_2080_352_360.nc
- 006_2006_2080_352_360.nc
- 007_2006_2080_352_360.nc
- 008_2006_2080_352_360.nc

6 Datasets with the same Spatial and Temporal Coverage

- 66 locations across UK (6 lon x 11 lat)
- 27,374 days (2 January 2006 - 12 December 2080)
- 9 variables: TREFHT, TREFMAX_U, FSNS, FLNS, PRECT, PRSN, QBOT, UBOT, VBOT (same as project 1)



What's different?
Slightly different values

Which means:
- Ensemble/multiple runs of the same simulation

The Provided Datasets:
Climate Data from Year 2006 - 2080

003_2006_2080_352_360.nc

004_2006_2080_352_360.nc

005_2006_2080_352_360.nc

006_2006_2080_352_360.nc

007_2006_2080_352_360.nc

008_2006_2080_352_360.nc

Merged Dataset
(averaged)

▸ Dimensions:	(**time**: 27374, **lat**: 11, **lon**: 6)		
▾ Coordinates:			
lat	(lat)	float32	49.48 50.42 51.36 ... 57.96 58.9
lon	(lon)	float32	352.5 353.8 355.0 356.2 357.5 358.8
time	(time)	float32	1.0 2.0 3.0 ... 2.737e+04 2.737e+04
▾ Data variables:			
TREFMXAV_U	(time, lat, lon)	float32	nan nan nan nan ... 283.3 nan nan
FLNS	(time, lat, lon)	float32	40.86 40.79 37.88 ... 65.62 63.85
FSNS	(time, lat, lon)	float32	30.44 25.66 25.82 ... 7.944 8.096
PRECT	(time, lat, lon)	float32	3.643e-08 4.111e-08 ... 4.56e-08
PRSN	(time, lat, lon)	float32	4.181e-16 3.793e-16 ... 3.379e-14
QBOT	(time, lat, lon)	float32	0.007071 0.006948 ... 0.00499
TREFHT	(time, lat, lon)	float32	285.2 284.9 284.5 ... 282.1 282.3
UBOT	(time, lat, lon)	float32	8.525 7.8 6.794 ... 3.484 4.03 5.01
VBOT	(time, lat, lon)	float32	3.041 3.064 3.134 ... 4.333 4.292
▾ Indexes:			
lat	PandasIndex		
lon	PandasIndex		
time	PandasIndex		
▸ Attributes: (0)			

Merge the Datasets: Take the Average

- Reduce random variability
- Enhance robustness
- Improve comparisons with external datasets which normally use multi-model ensembles (averaging across models)

The External Dataset:

One dataset containing:

- Time
- Latitude
- Longitude
- 2-metre air temperature (t2m)
- Soil temperature at layer 1 (stl1)
- Volumetric Soil Water at layer 1 (swvl1)

- 6,940 timetamps
- 39 latitudal coordinates
- 26 longitudal coordinates
- Time ranging from 01/01/06 to 31/12/24
- Latitude ranging from 49.4 to 58.9
- Longitude ranging from -7.5 to -1.25

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Data Preprocessing of the Provided Datasets

5 steps taken to ensure dataset is ready for further modelling

1 Correct Datatype

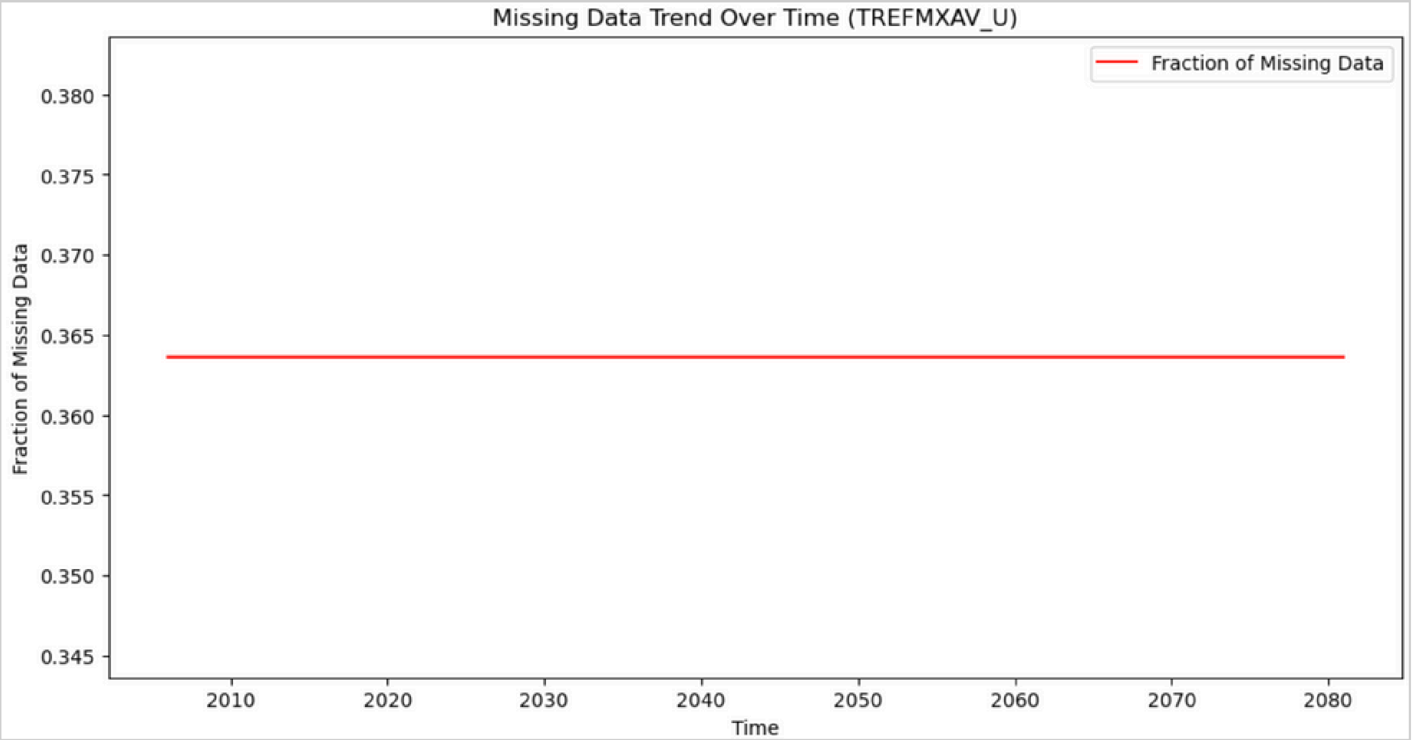
- "Time" from float32 to datetime

2 Duplicates

- No duplicated values found

3 Missing Values

- 656,976 missing values in TREFMAX_U (36.4%)
- Check spread of MVs



- Check gaps between MVs

```
{'Total Gaps': 218993,
 'Min Gap Size': np.int64(1),
 'Max Gap Size': np.int64(7),
 'Mean Gap Size': np.float64(2.9999863009319934),
 'Median Gap Size': np.float64(2.0)}
```

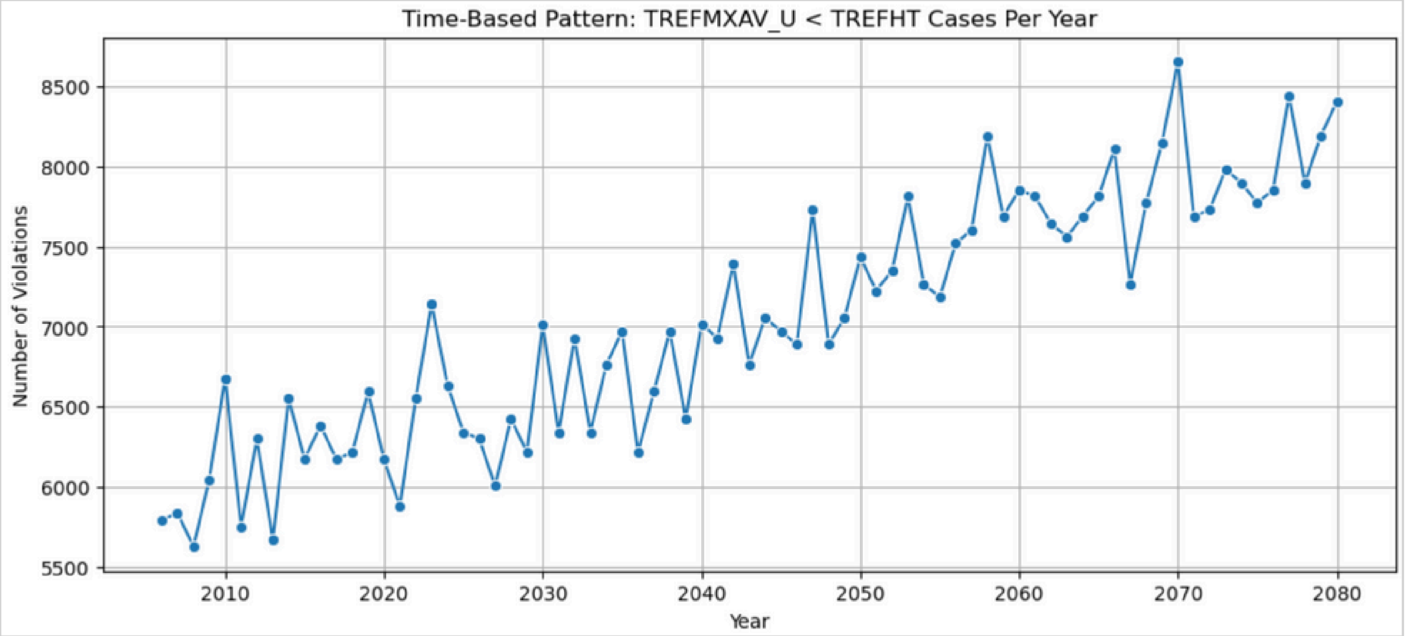
- % MVs remains constant overtime (randomly distributed).
- Gap size is relatively small (2-3 steps missing at a time)
- Use forward fill to impute MVs

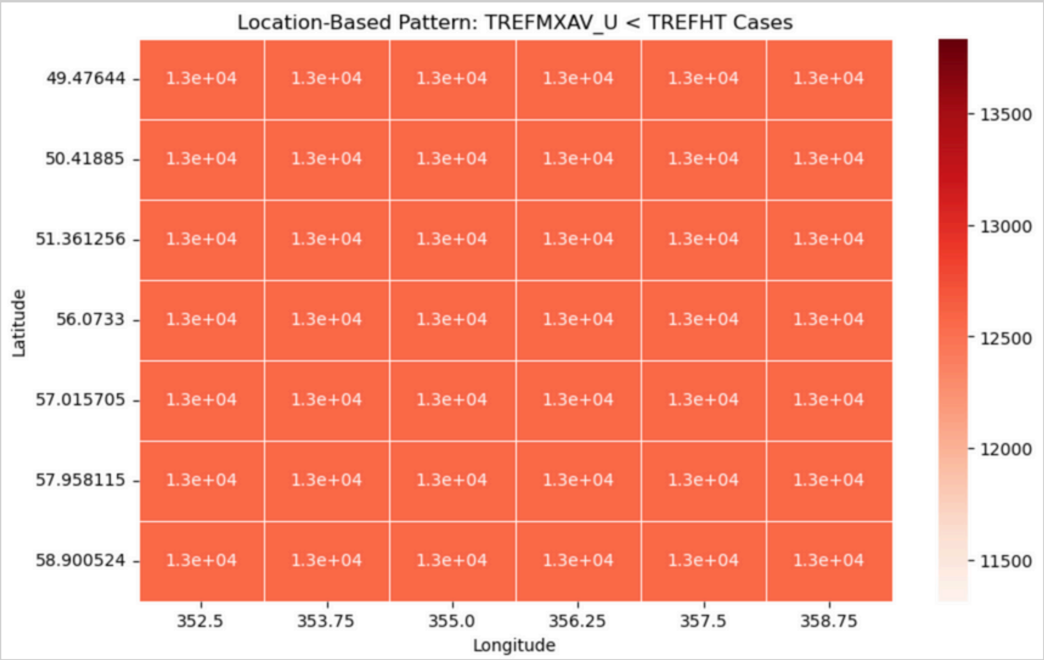
4 Adjust Units

- TREFHT and TREFMAX_U from Kelvin to Celcius

5 Fix Incorrect Values

- Replace 45,477 negative values in PRSN with 0
- Handle 196,499 cases where TREFMAX_U is lower than TREFHT
  - Check distribution of error (against time and location)

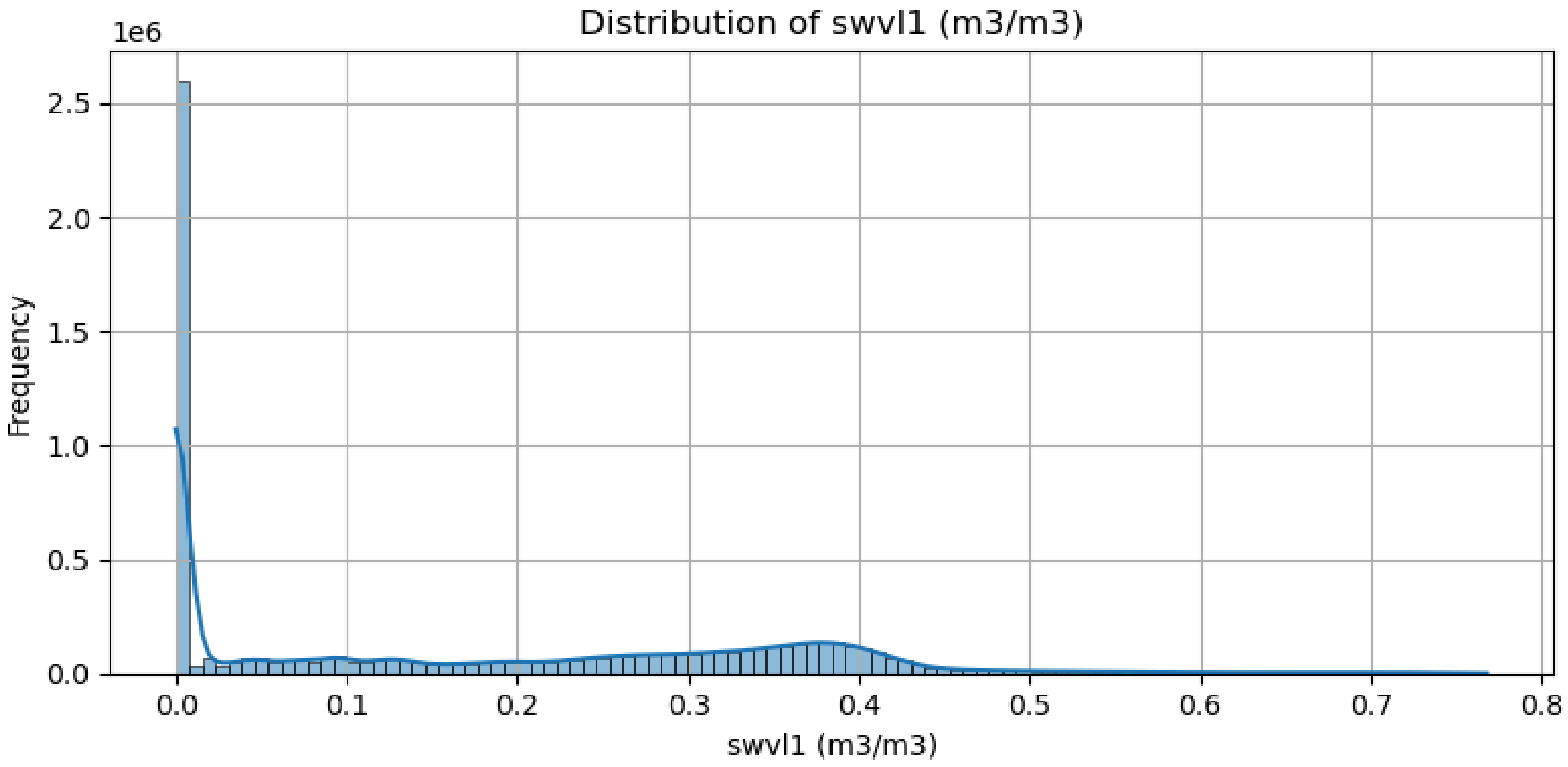




- Cases where TREFMAX_U < TREFHT occur consistently over time with a slight increasing trend
- Also distributed evenly across locations
- Systemic error

- Divide errors into 2:
  - Small error (0 to 1°C) --> setting TREFMAX_U = TREFHT + 0.1°C
  - Large error (> 1°C) --> median-based correction with scaling

Data Preprocessing of the External Datasets



- Align the component and naming of coordinates (time, lat, lon)
- Set time to index
- Check for missing values and duplicates
- Convert temperature from Kelvin to Celsius
- Check negative values

- Negative values in swvl1 skewed the distribution of the data and needed to be investigated further as negative values are invalid.
- Negative values imputed with the median as it is robust to skewness and outliers unlike the mean.

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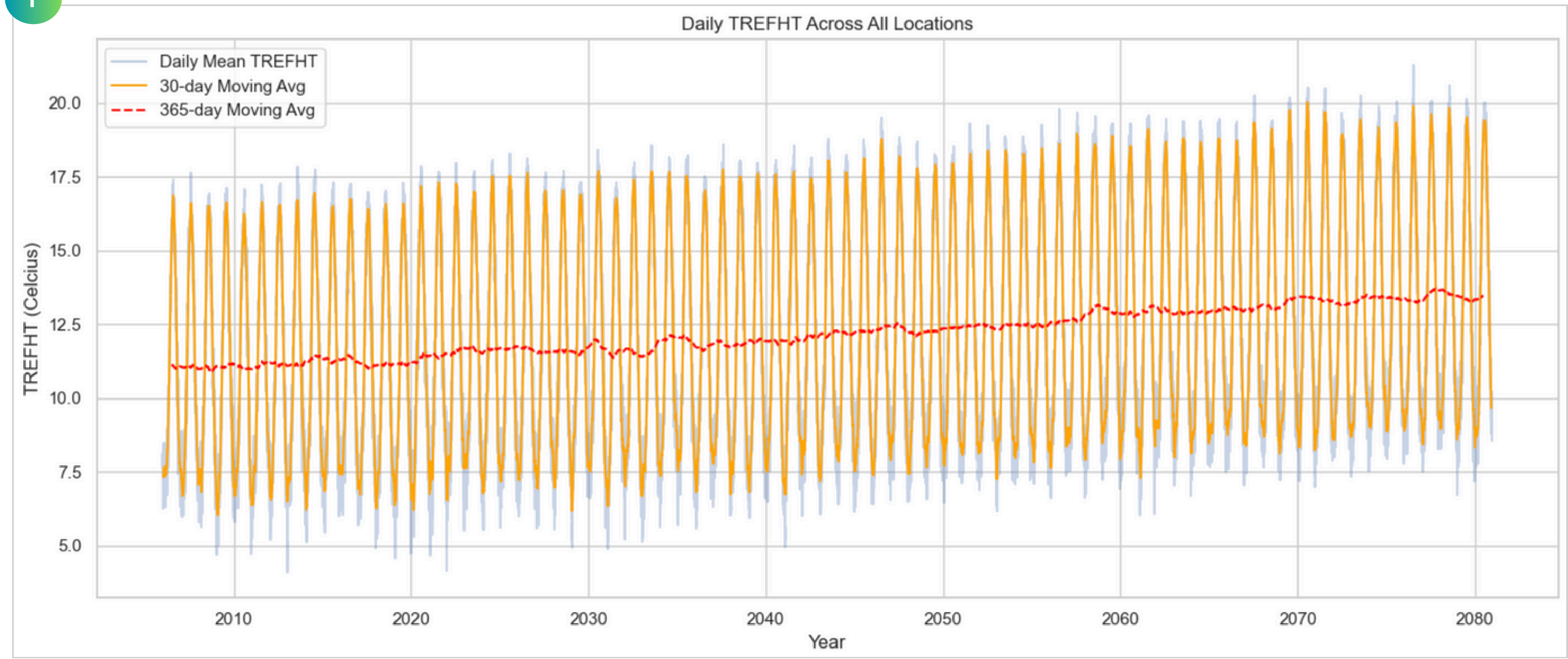
Temporal Analysis of TREFHT

A visual summary of long-term temperature trends, seasonal cycles, and projected climate changes using TREFHT data

1



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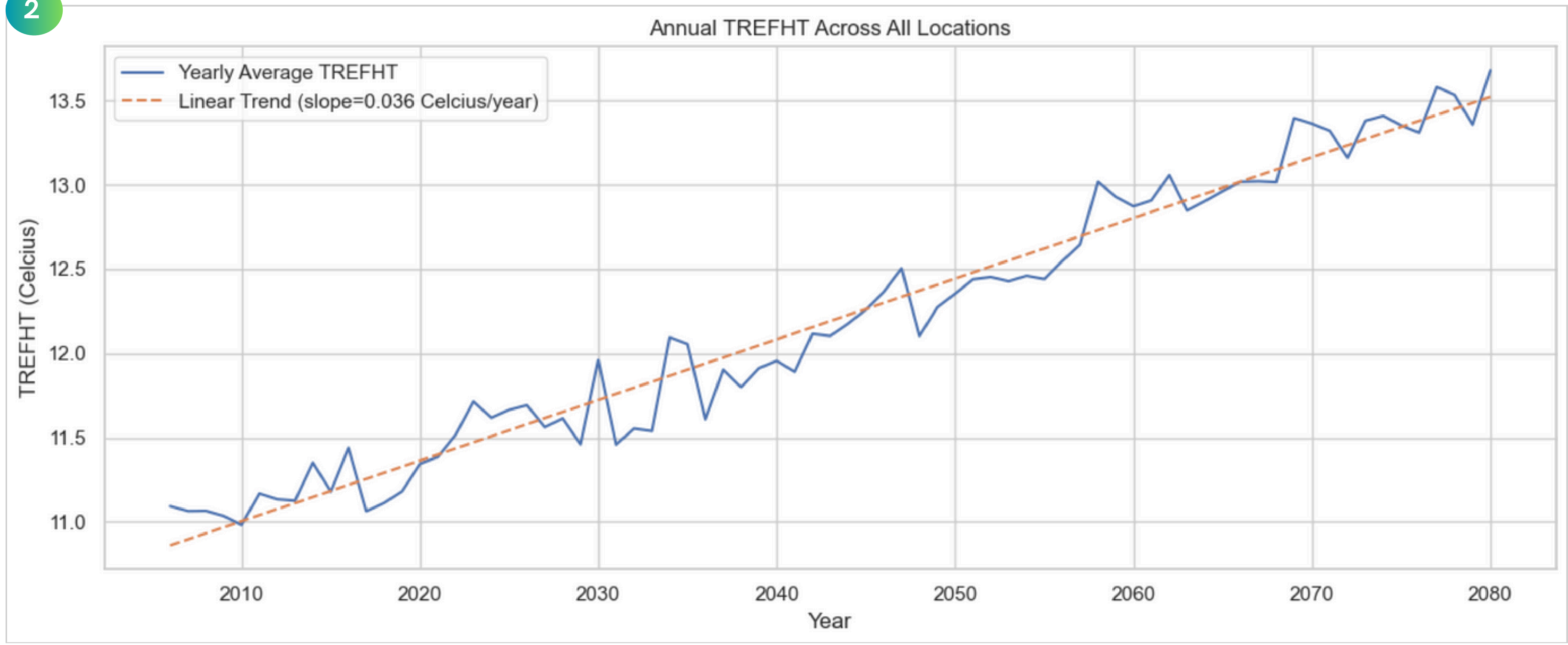

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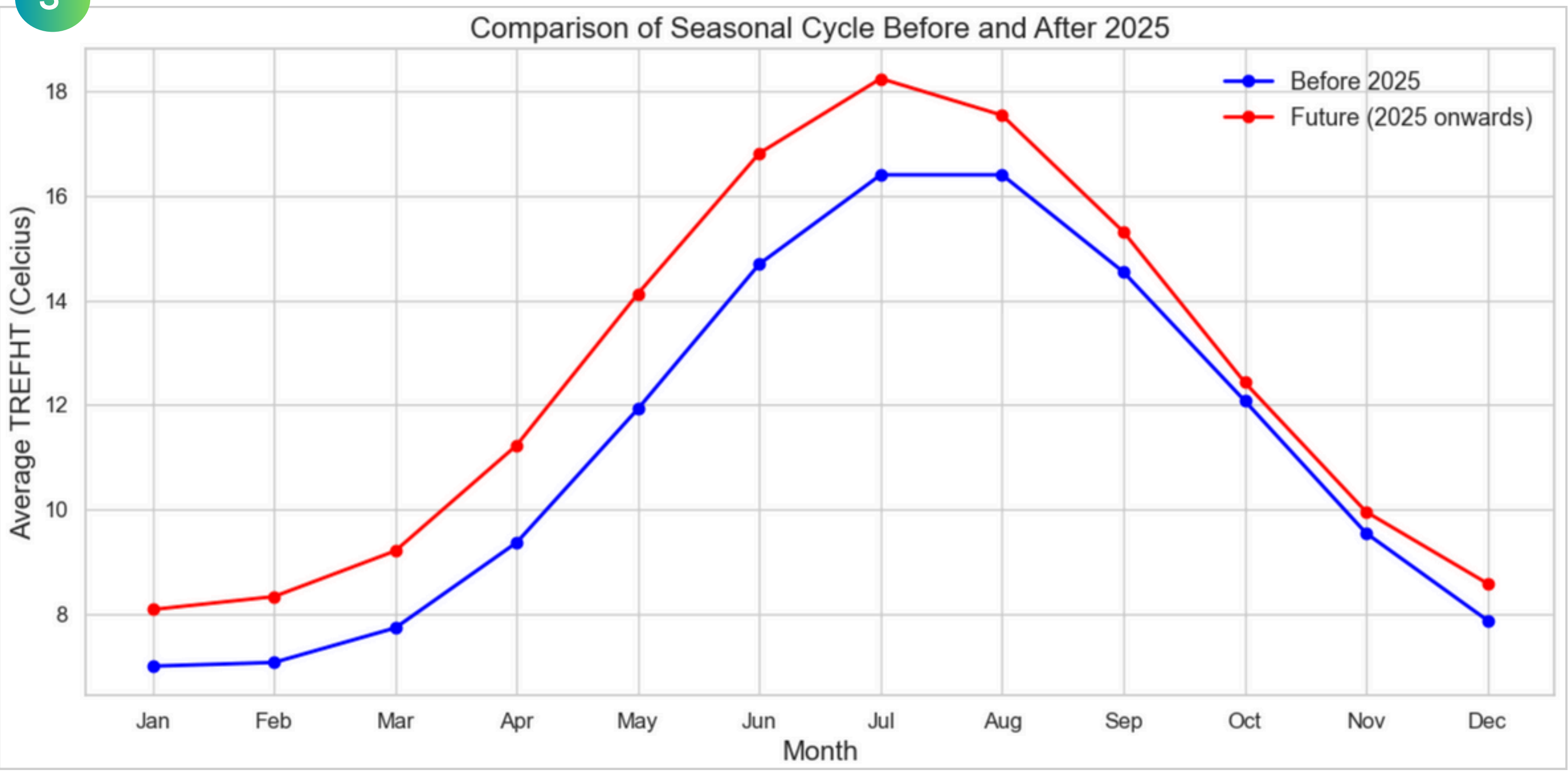


Key Takeaways

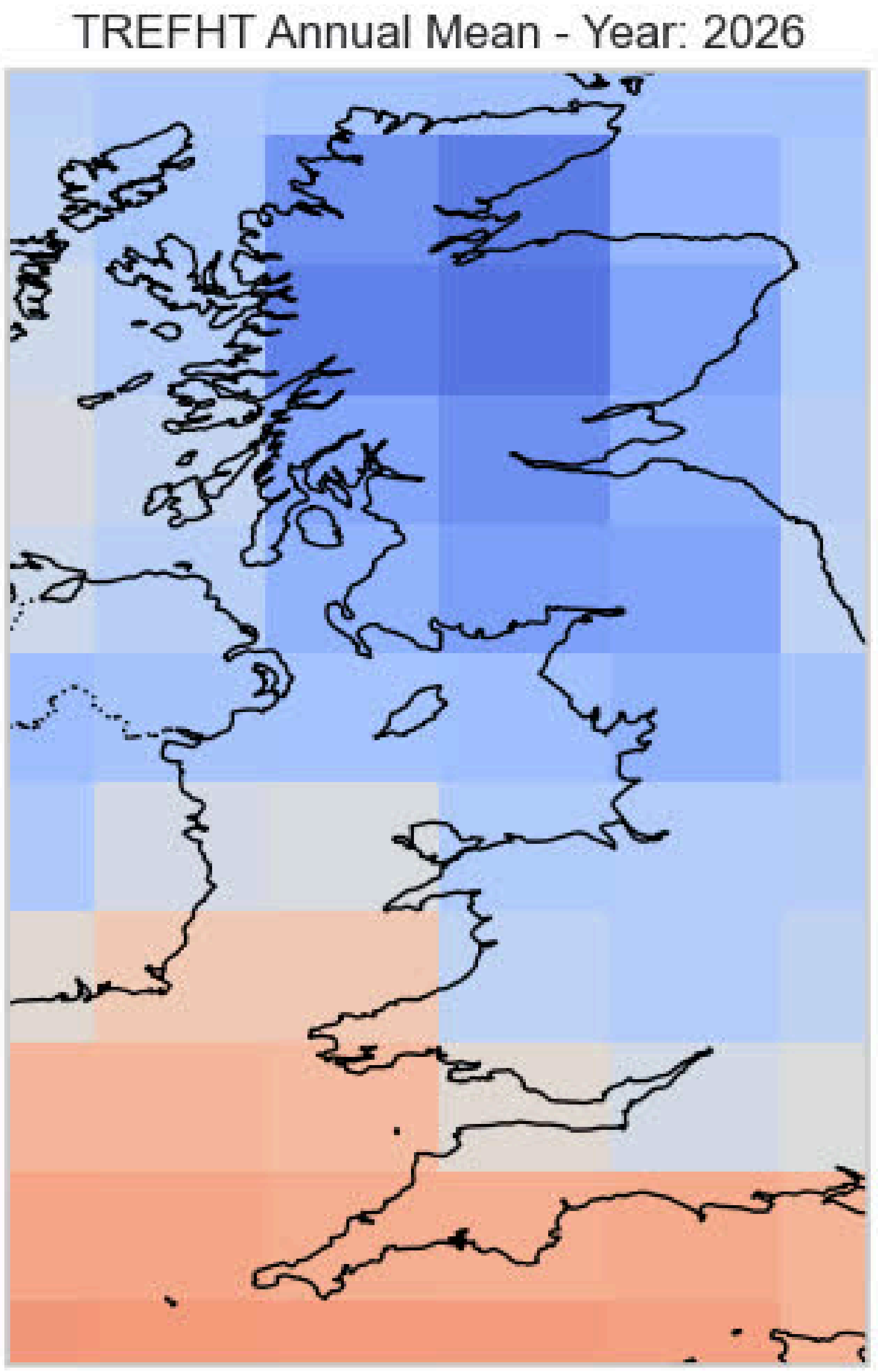
(1) The consistent annual cycle and rising long-term trend suggest climate change impacts. Seasonal extremes (peaks and throughs) also appear to grow over time (summer vs winter temperature gap becomes larger).

(2) Annual mean TREFHT with a linear trend line, reinforcing the overall warming despite natural yearly fluctuations.

(3) The red line shows consistently higher temperatures than the blue, confirming a warming temp in the future. Seasonal extremes grow more pronounced, with future summers ~12.5% hotter and milder winters. Sharper transitions suggest shorter spring and autumn periods.

Spatiotemporal Analysis of TREFHT

A visual overview of how near-surface air temperature (TREFHT) changes across space and time in the UK.



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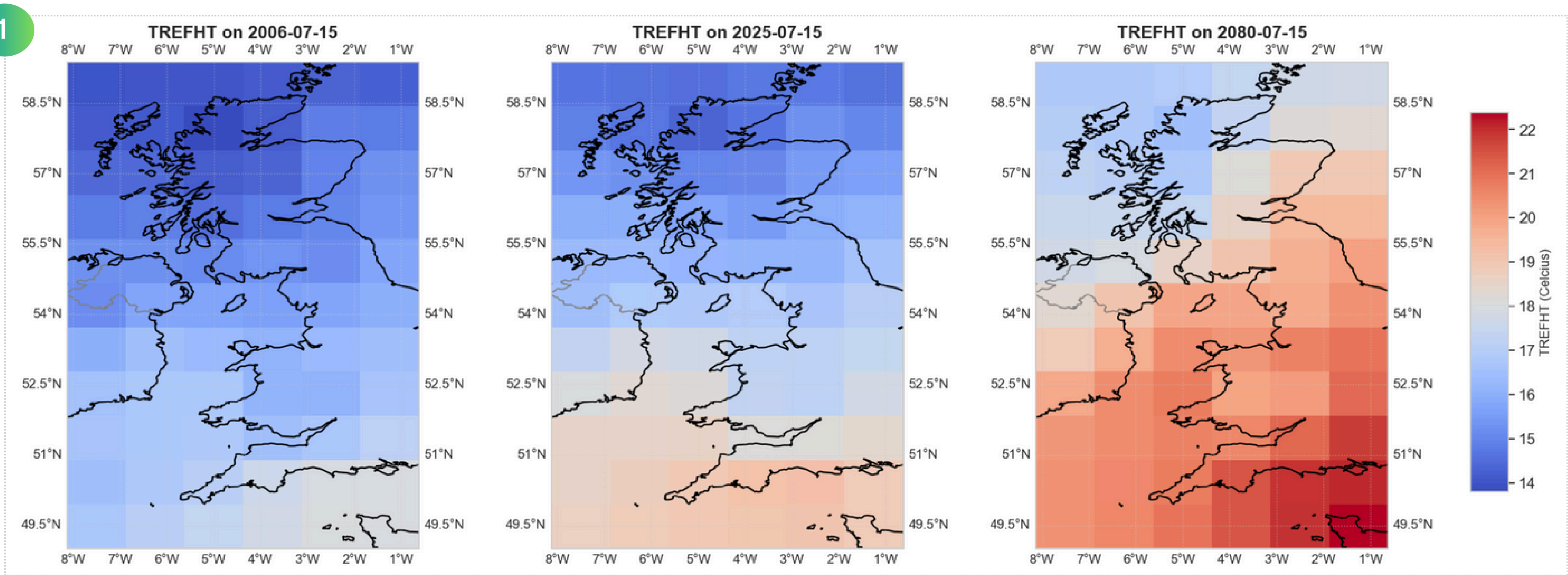

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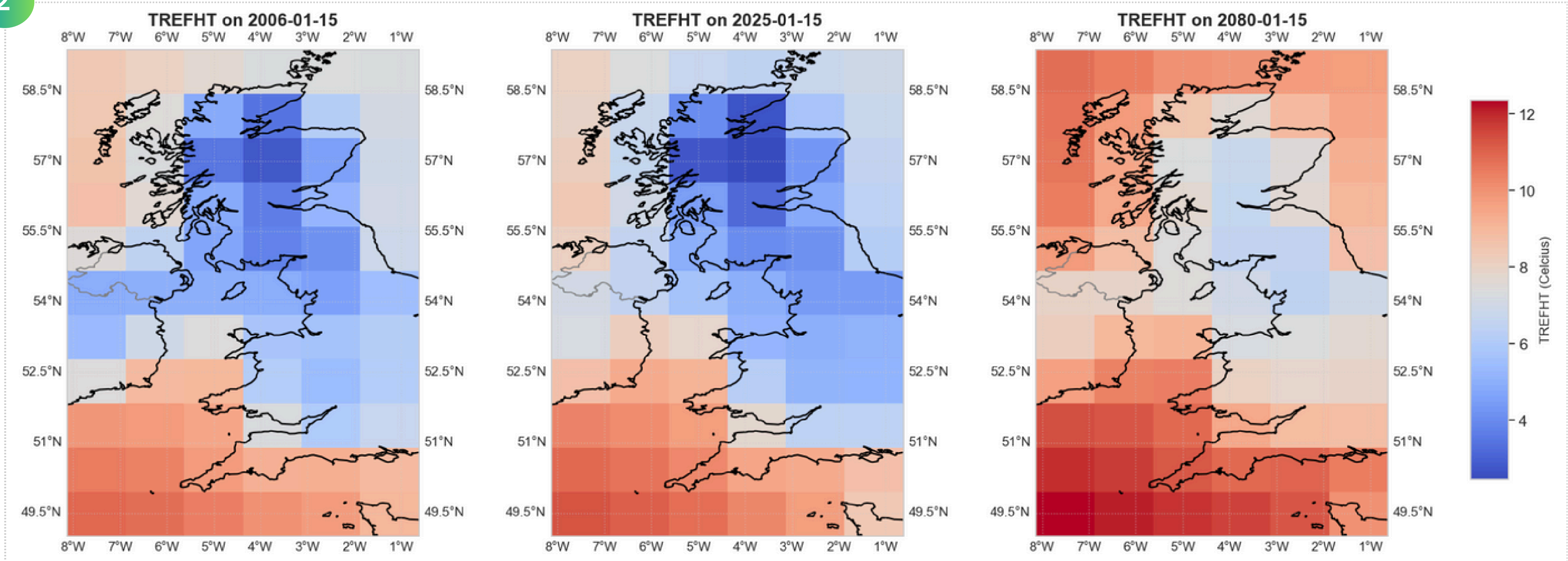


Key Takeaways

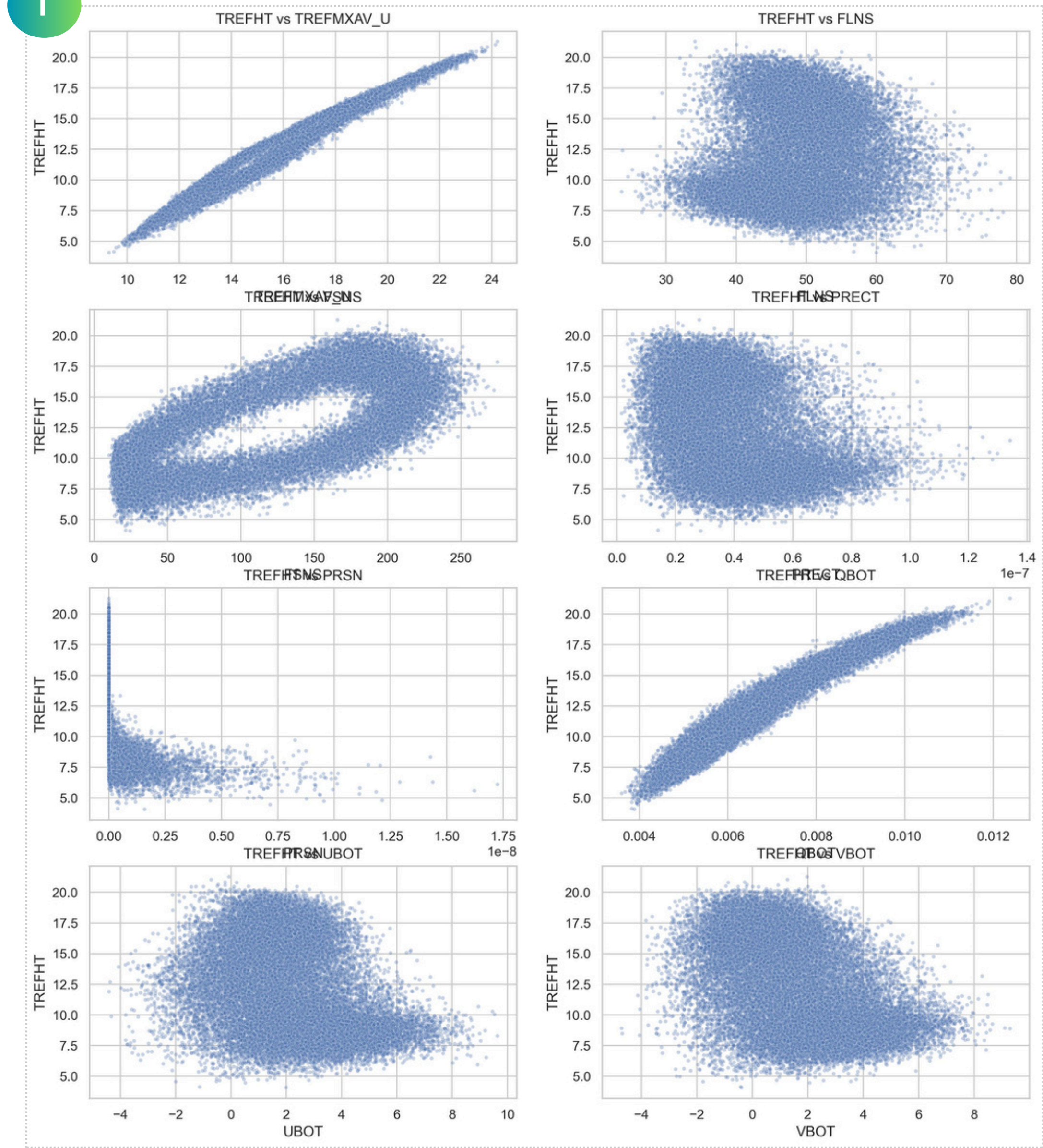
(1) Temperatures in the UK on 15 July (peak summer) across 2006, 2025, and 2080, with a clear warming trend—especially in southern England, where temperatures exceed 22°C by 2080.

(2) Temperatures across UK on 15 January (peak winter) from 2006 to 2080, with much of England exceeding 10°C by 2080—indicating a clear warming trend and milder winters.

Understanding TREFHT Drivers: Correlation and Relationship Patterns

This slide explores how TREFHT relates to key climate variables, highlighting both overall correlations and regional variations across the UK.

1



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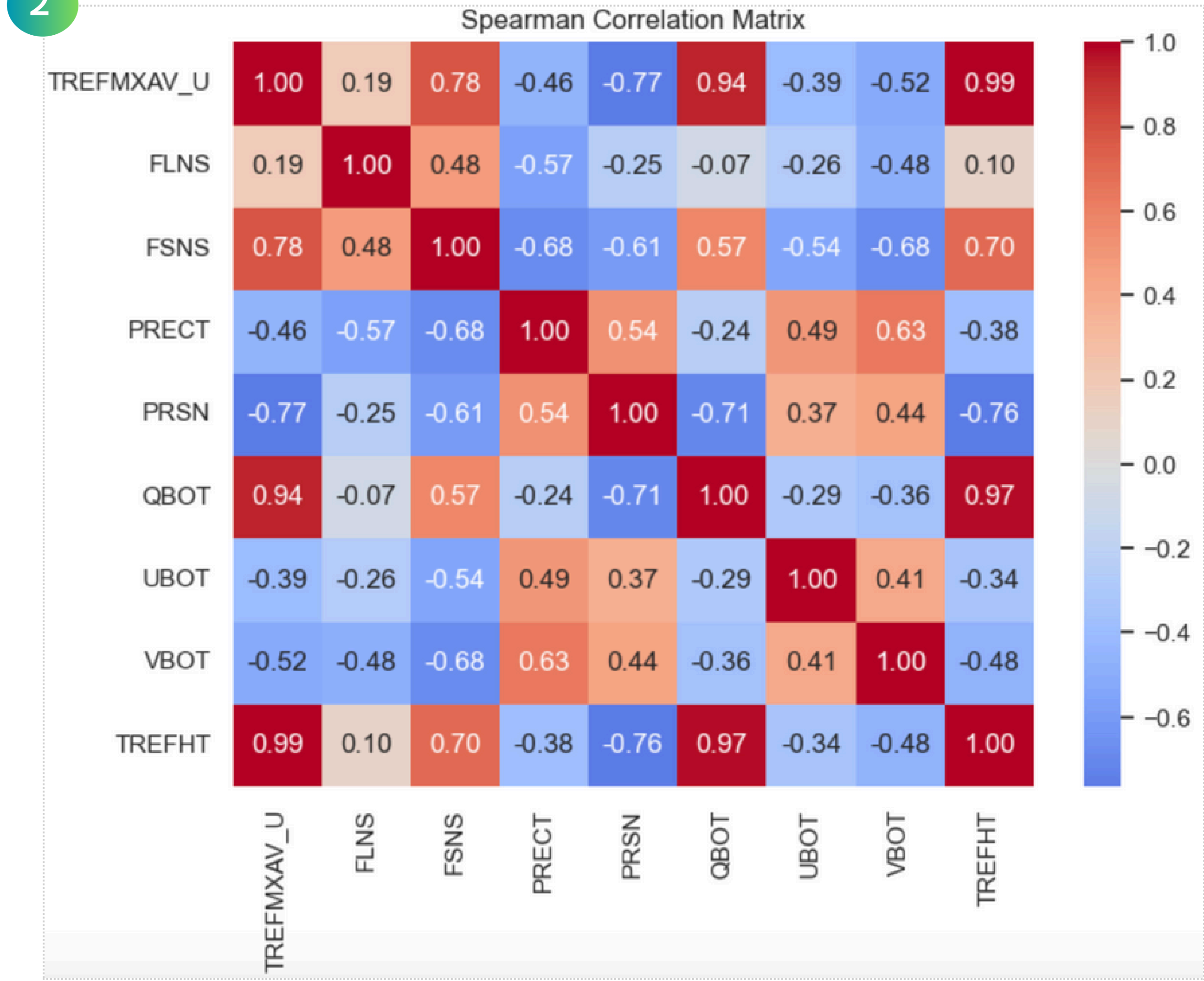

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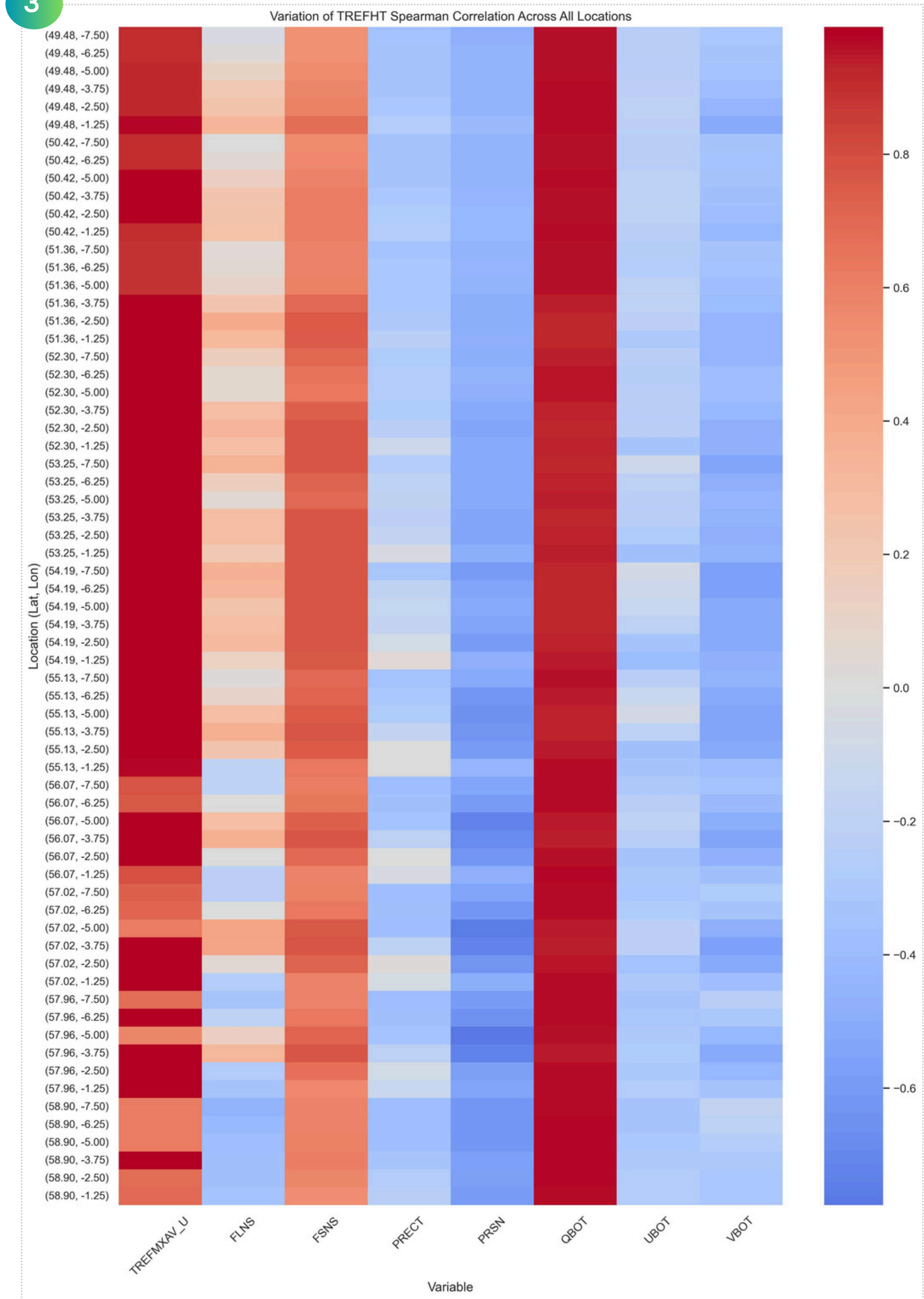


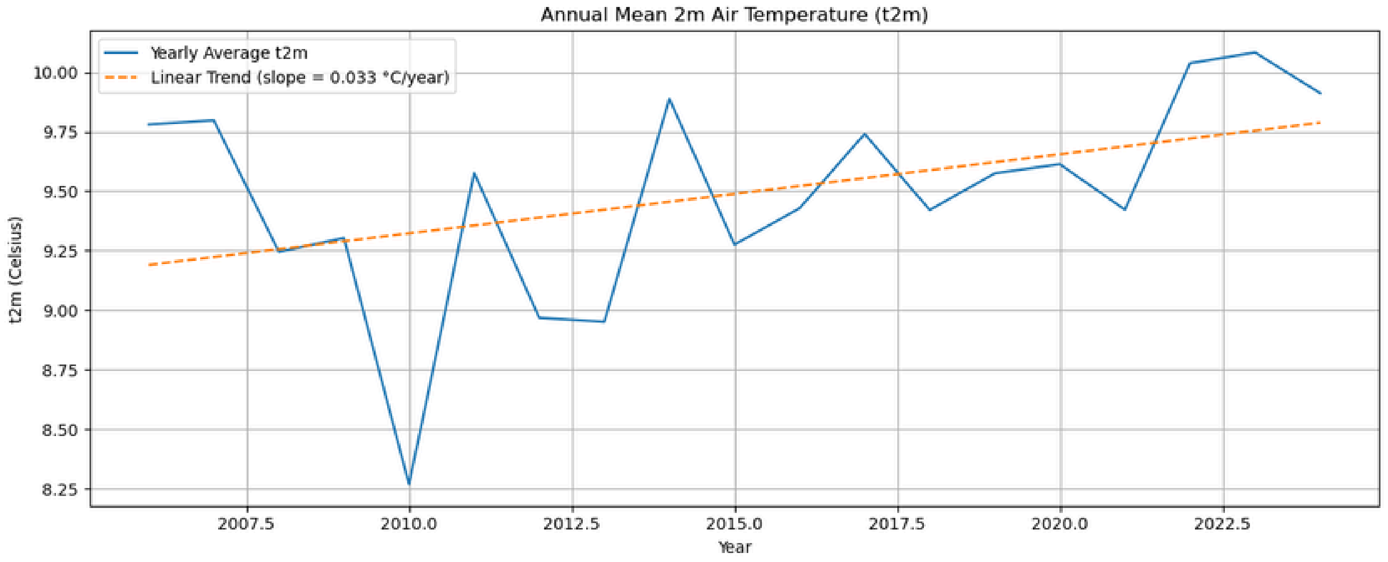
Key Takeaways

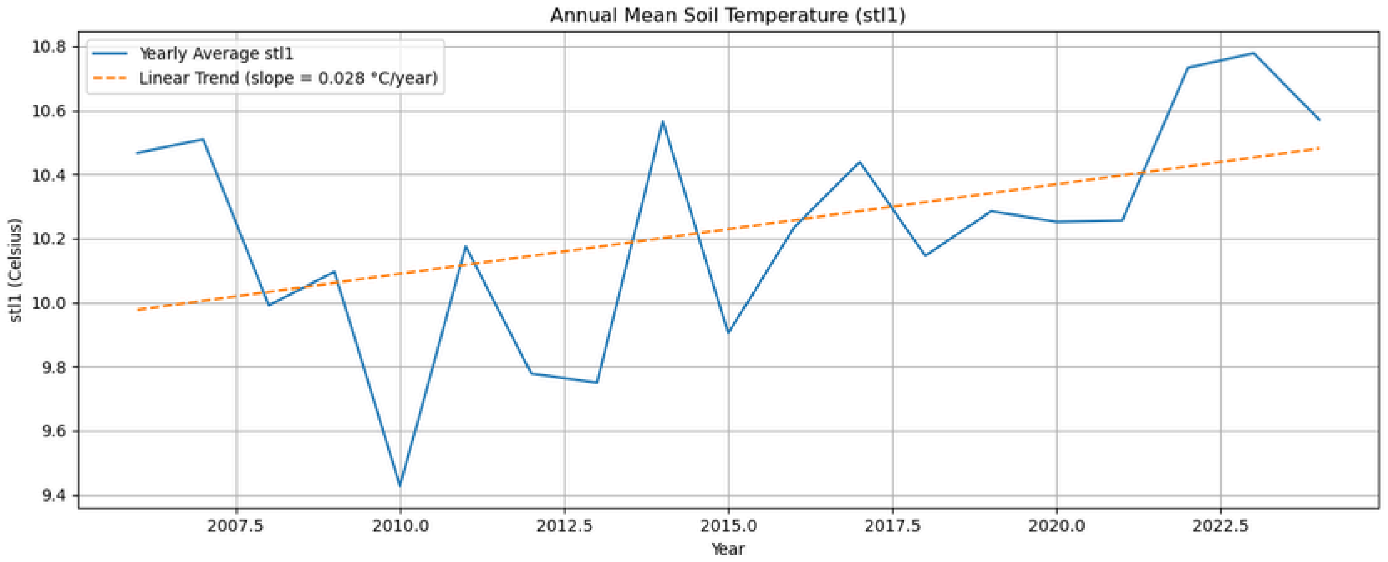
(1) TREFHT Shows Mixed Linear and Nonlinear Relationships with Predictors

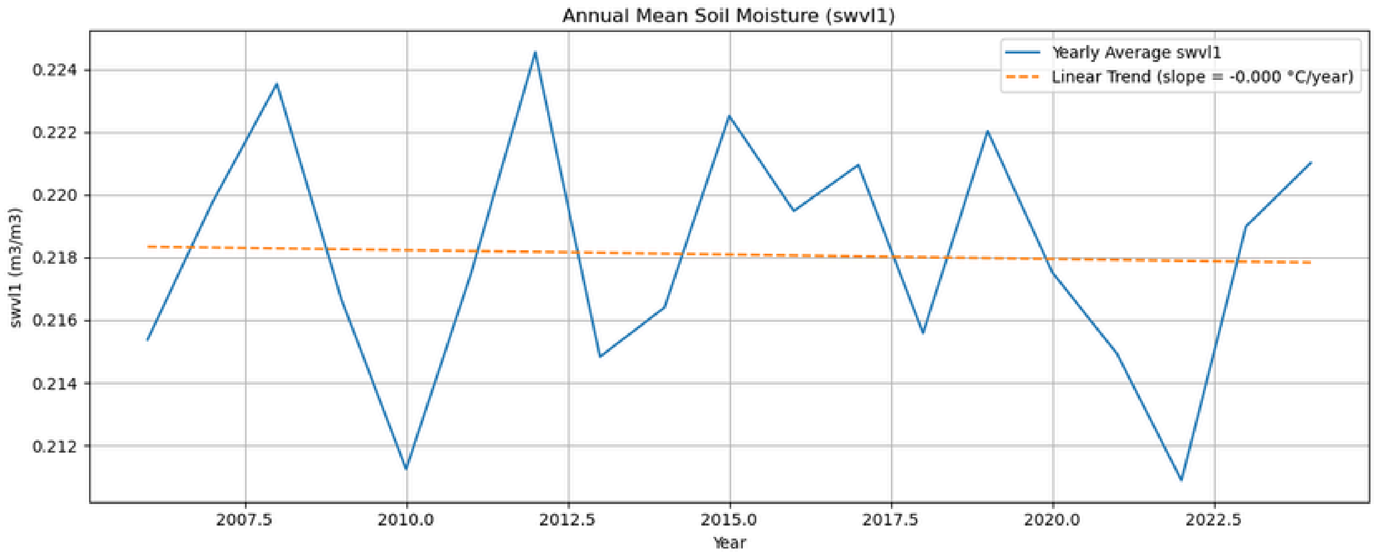
(2) QBOT/humidity (ρ = 0.97), PRSN/snowfall rate (ρ = -0.76), and FSNS/solar flux (ρ = 0.70) have the strongest correlation with TREFHT

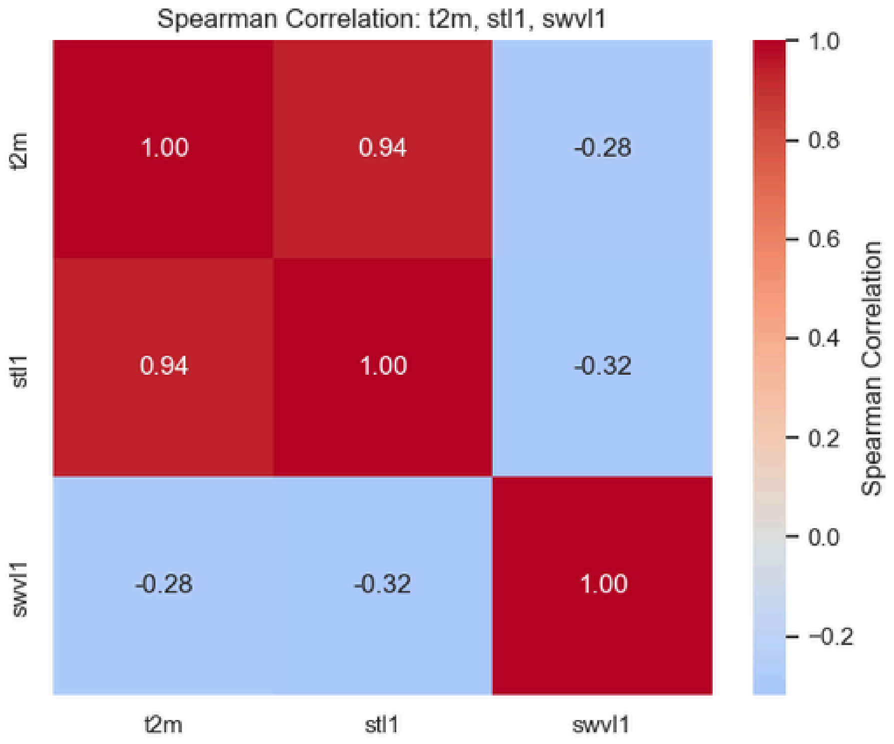
(3) Compare spearman correlation matrix across all locations to see if the key factors are consistent: QBOT, FSNS, and PRSN consistently correlate strongly with TREFHT across locations, while FLNS and wind variables vary more by region. This suggests core drivers are stable, but some effects are location-dependent.

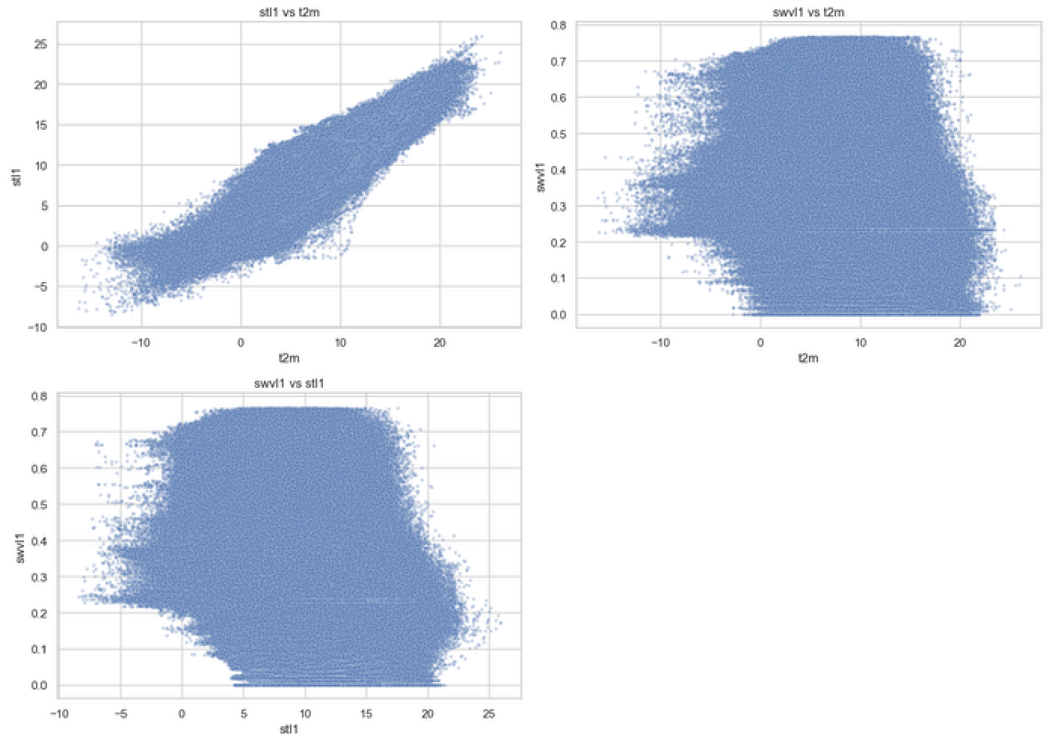
Understanding Topsoil Temperature & Moisture: Correlations & Trends











- Annual mean soil and air temperature show simiar trends over time with a steady increase between 2006-2024 at a similar rate (approx. 0.03 clesius/year).
- These two variables show a strong positive correlation
- As temperatures have risen, soil moisture has slowly decreased, showing a weak-to-moderate negative correlation with both temperature variables
- Soil moisture trending towards 0.2 m^3/m^3 which would be where plants begin to experience water stress, especially during dry periods
- Should water moisture drop below 0.1, the soil will become very dry, posing serious risks such as drought and crop failure

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Modelling Future Soil Moisture under Climate Change

A Two-Stage Framework Combining Time Series Forecasting and Machine Learning

Stage 1

CEMS Datasets (2006-2080)

Variables

- Target: TREFHT
- Predictor: FSNS, FLNS, PRECT, PRSN, QBOT, UBOT, and VBOT

Train vs Test

- Train: 2006-2024
- Test: 2025-2080

Methodology

Three modelling strategies were explored for temperature forecasting:
(1) Prophet - applied to the average TREFHT across all spatial locations
(2) XGBoost - using the average TREFHT values across locations
(3) XGBoost with XESMF interpolation - preserving the spatial component to enable location-specific predictions

Each model was evaluated using cross-validation to improve generalisation and ensure robust performance.

Output

Projected TREFHT 2025-2080

Use this projection to validate the synthesis TREFHT data in the given dataset, choose the projection with the lowest error, then use it as input for the second modelling stage.

Stage 2

Projected TREFHT from stage 1

ERA 5 Soil Dataset

Variables

- Target: swvl1 (soil moisture)
- Predictor: t2m (temperature)

Train vs Test

- Train: 2006-2024
- Test: 2025-2080

Methodology

An XGBoost regression model was trained to predict future soil moisture levels. This approach allowed us to examine how projected temperature increases may influence surface soil moisture, which is critical for agricultural productivity and land surface processes.

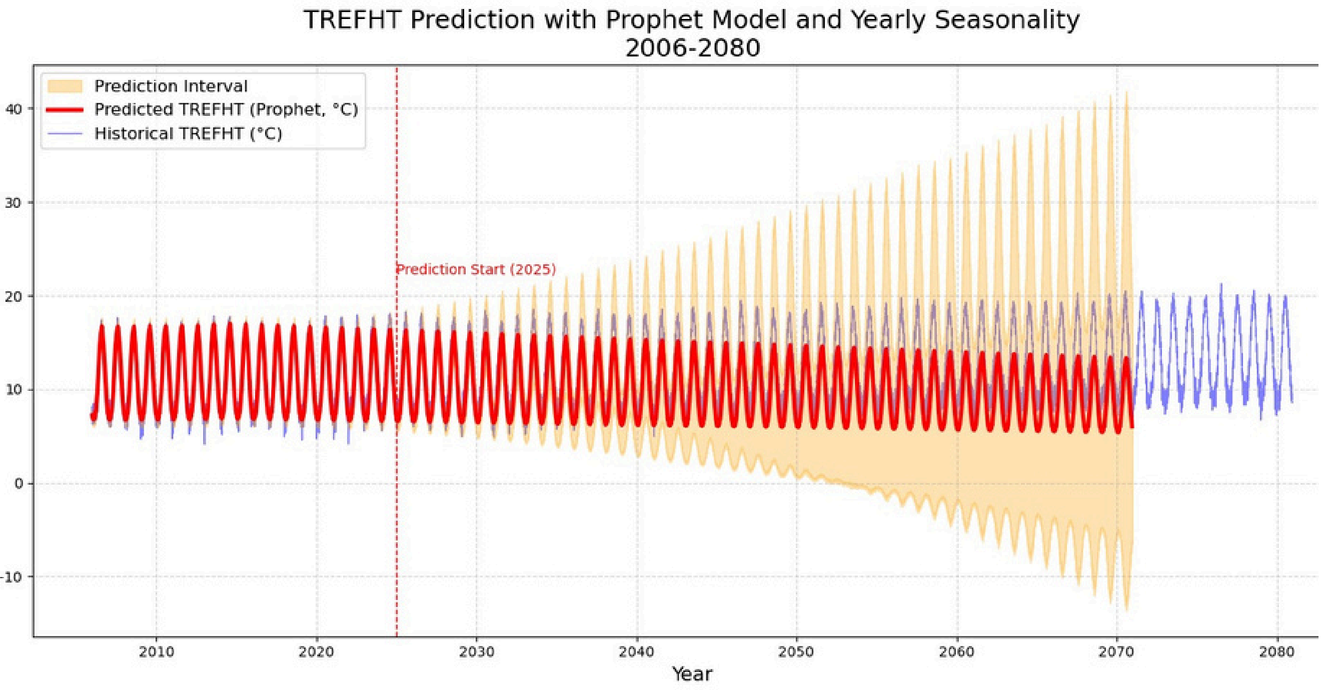
Output

Projected soil moisture 2025-2080

Compare this projection with the soil moisture standard from Food and Agriculture Organization (FAO) to see how rising temperature impacts future crop yields

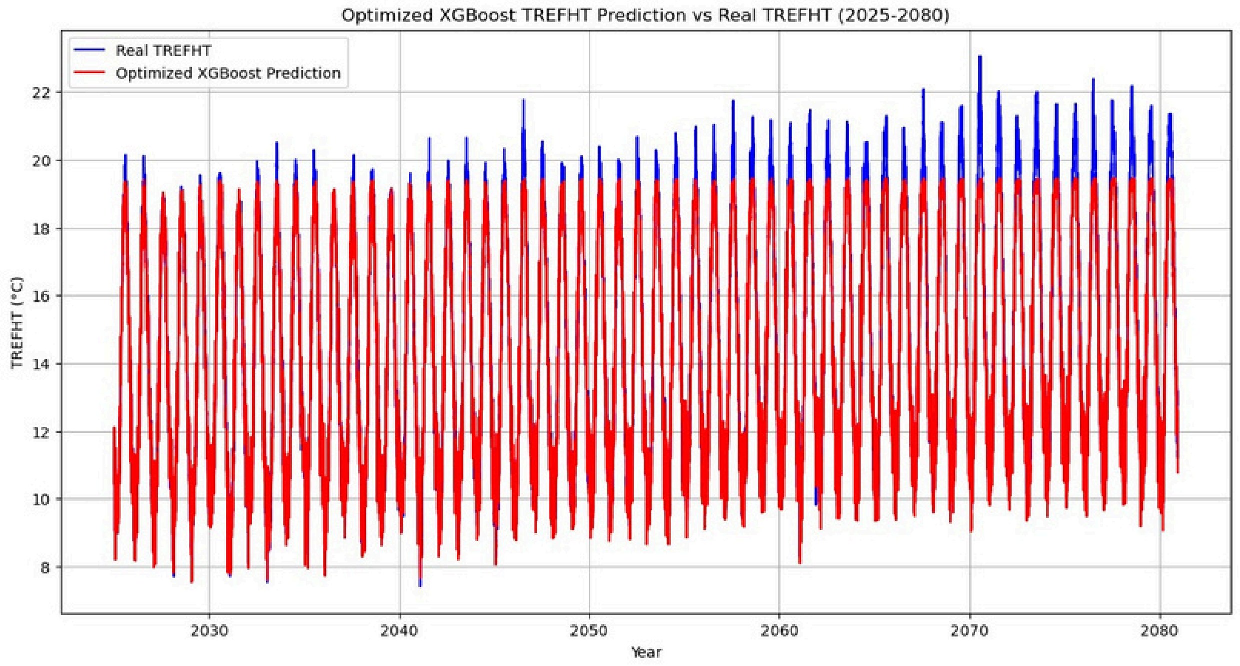
Forecasting Temperature with 3 models

Modelling Long-Term Trends in Near-Surface Air Temperature (TREFHT)



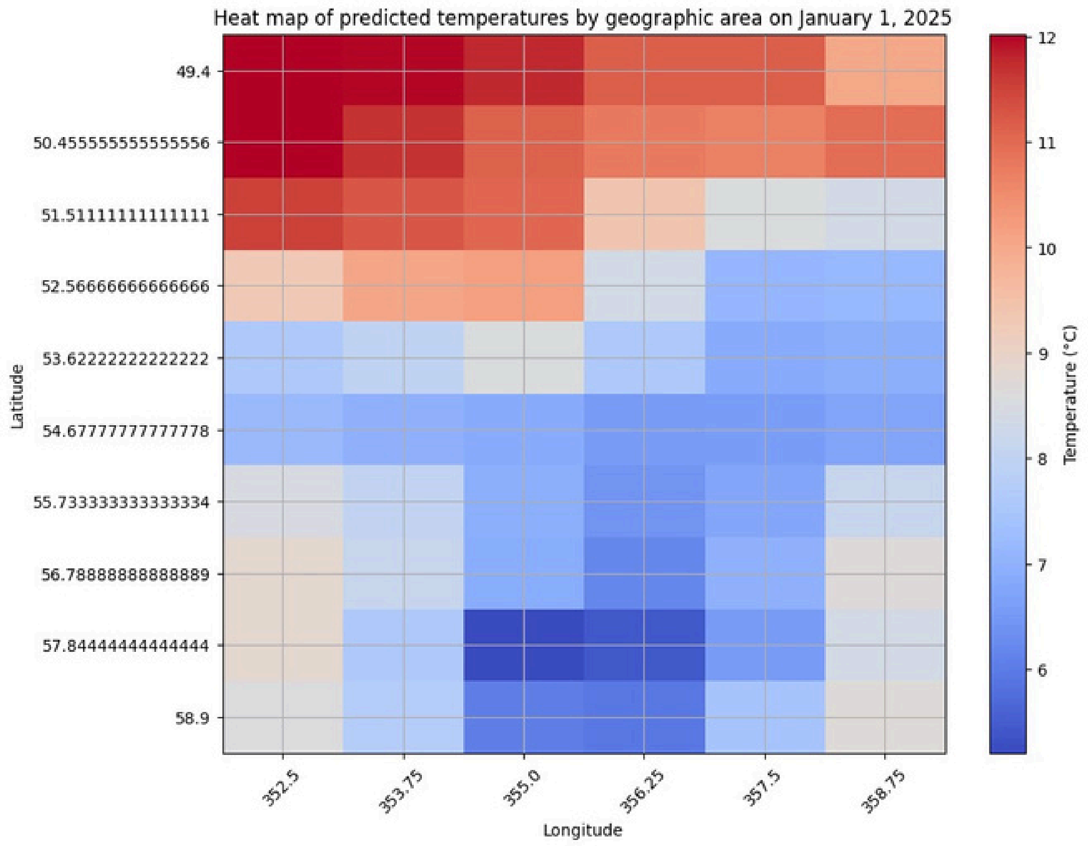
Prophet Forecast

Forecasted average TREFHT with Prophet model. Shows seasonality but weak long-term warming trend. Prediction uncertainty increases sharply over time.



XGBoost Forecast (per location)

Time series prediction at one grid point (lat 50.42, lon 355.0). Captures seasonal cycles and upward trend more clearly than Prophet.



Heatmap from XESMF + XGBoost

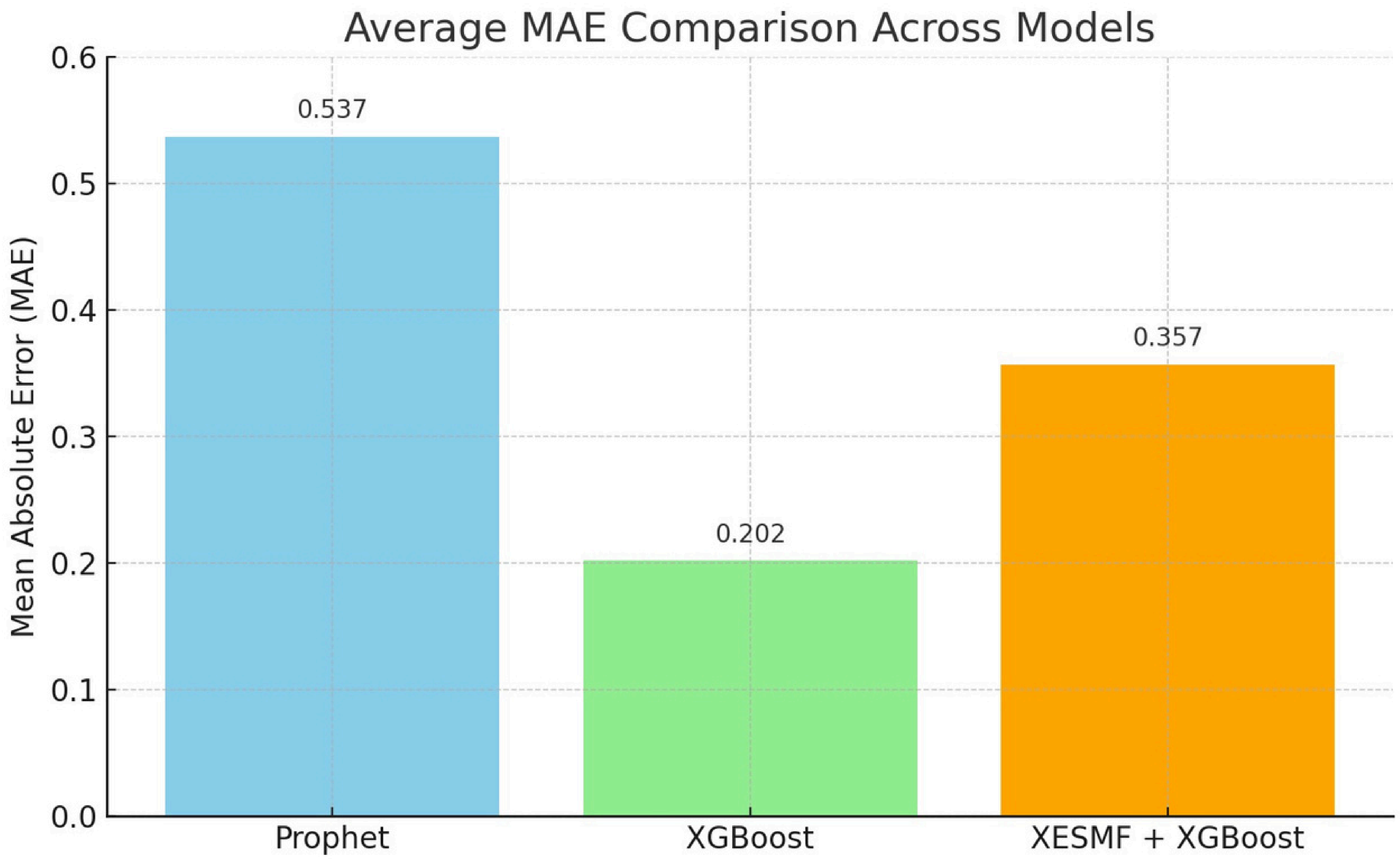
Spatial distribution of predicted TREFHT on Jan 1, 2025. Shows geographic variability in near-surface temperature across the region.

Comparing Different Models

Choosing the Best Model with the Lowest Error

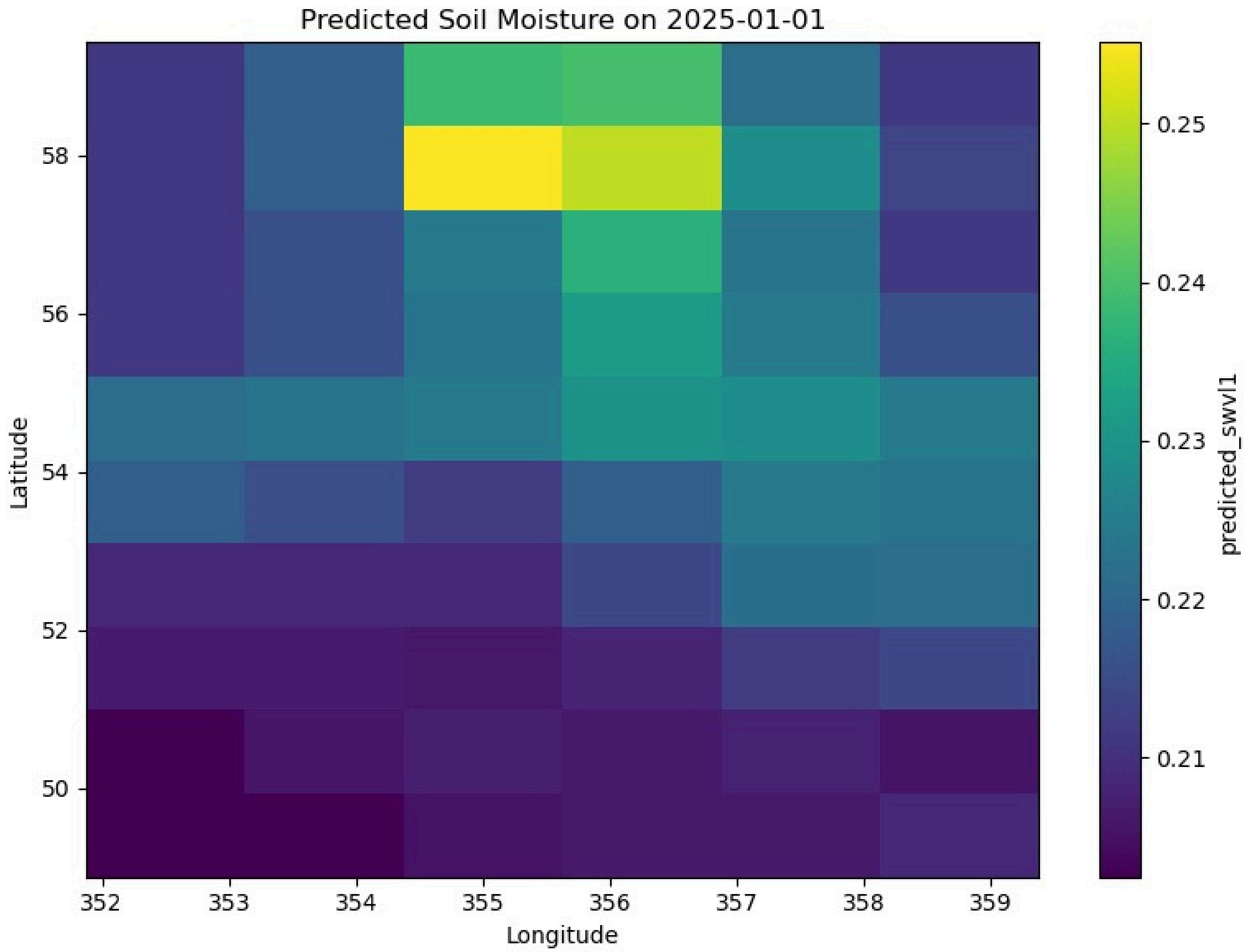
Key Takeaways

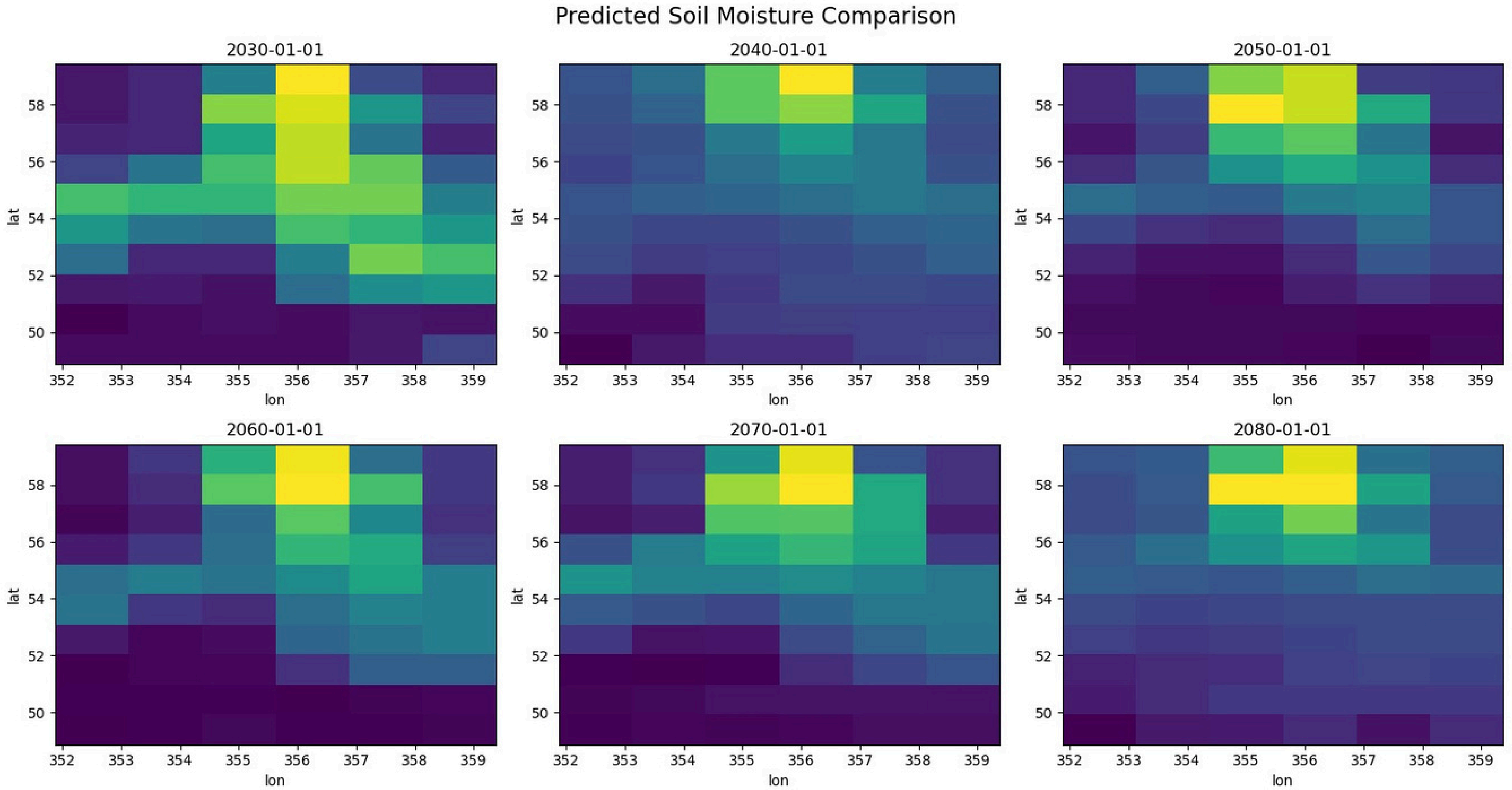
- **Prophet** has the highest error, indicating it struggles with capturing long-term trends accurately
- **XGBoost** performs the best with the lowest error, making it well-suited for modeling complex nonlinear patterns.
- **XESMF + XGBoost** falls in between in terms of accuracy, possibly trading off some precision for better spatial consistency.

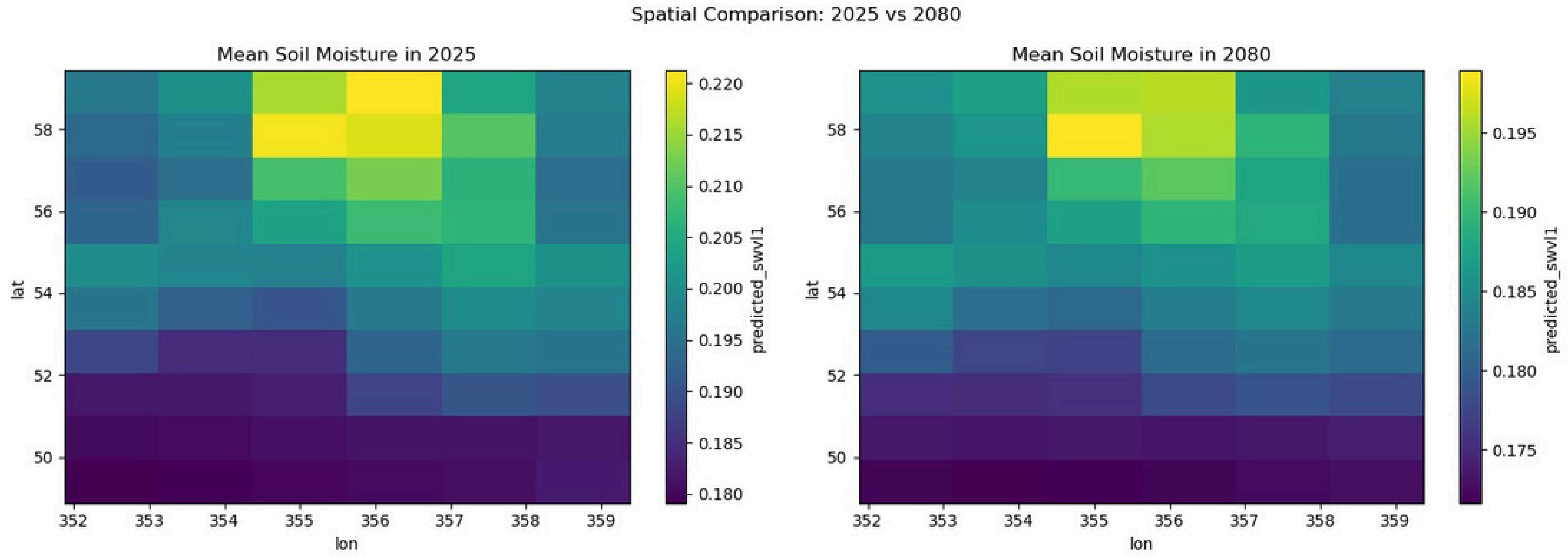


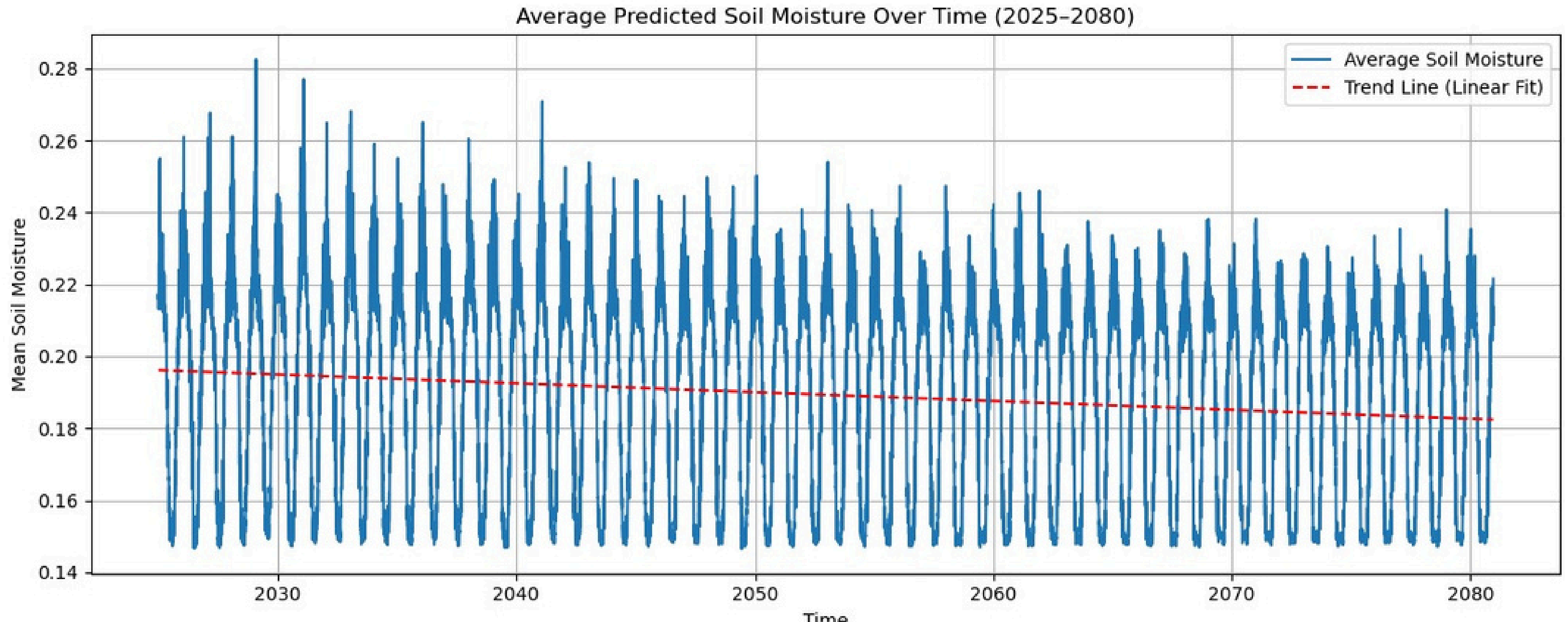
Projection of Soil Moisture 2025-2080

Use the Predicted TREFHT for Soil Moisture Projection









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Result Interpretations: Climate Impacts on Soil Moisture and Agriculture

From Rising Temperatures to Drying Soils: Implications for Crop Health and Ecosystem Stability

1 Temperature

✅ TRFHT is projected to rise steadily through 2080 across all areas in the UK

2 Soil Moisture

✅ The rising temperature is closely linked to the decline in soil moisture from 0.218 in 2006 to a projection of 0.18 in 2080

✅ Seasonal pattern persists until 2080: higher moisture in summer, but the overall trend is decreasing

3 Agricultural Impact

✅ Soil Moisture Benchmarks (FAO):

- 🌱 0.3–0.4 m^3/m^3 → Field capacity (ideal for crops, balanced water cycle)
- 🌤️ 0.2 m^3/m^3 → Below field capacity (risk of water stress)
- 🔻 0.1 m^3/m^3 → Approaching wilting point (plants can't extract water)
- 🛑 0.1 m^3/m^3 → Severe drought stress, crop failure, and land degradation

✅ Agricultural & Environmental Impact

- ⚠️ Crop yields may decline, increasing irrigation demands.
- 💧 Reduced groundwater recharge and potential hydrological imbalance.
- 🌿 Ecosystem vulnerability through vegetation loss, erosion, and soil degradation.
- 🔁 Feedback loop: Dryer soils intensify heatwaves, escalating climate impacts (Saxton et al., 2006).

Agenda

Recommendations for Further Research

- Consider more predictors for the model when predicting soil moisture (e.g. top soil temperature) when modelling future predictio
- Remove the winter months before rerunning th model to focus on crop-growing seasons.
- How might rising temperatures affect other industries?
- Expand the area of analysis to a wider range geographical coordinates.

References

- D. Hillel, 2005. Thermal Properties And Processes. In: Encyclopedia of Soils in the Environment. Elsevier. Pages: 156-163. ISBN 9780123485304. https://doi.org/10.1016/B0-12-348530-4/00523-3.
- European Centre for Medium-Range Weather Forecasts, no date. ERA5 climate reanalysis data documentation. [online] ECMWF. Available at: https://confluence.ecmwf.int/pages/viewpage.action?pageId=197702790 [Accessed 25 Mar. 2025].
- European Environment Agency, 2021. Soil moisture deficit. [online]. Available at: https://www.eea.europa.eu/en/analysis/indicators/soil-moisture-deficit?activeAccordion=ecdb3bcf-bbe9-4978-b5cf-0b136399d9f8.
- IPCC, 2013. Summary for Policymakers. In: Stocker, T.F., Qin, D., Plattner, G.-K., Tignor, M., Allen, S.K., Boschung, J., Nauels, A., Xia, Y., Bex, V. and Midgley, P.M. (eds.) Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge: Cambridge University Press.
- IPCC, 2021. Summary for Policymakers. In: Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press. https://www.ipcc.ch/report/ar6/wg1/.
- Qian, B., De Jong, R., Gameda, S., Huffman, T., & Neilsen, D. (2010). Temporal and spatial patterns of soil temperature in Canada. Agriculture, Ecosystems & Environment, 138(3-4), 373-382. https://doi.org/10.1016/j.agee.2010.05.013
- Saxton, K. E., & Rawls, W. J. (2006). Soil water characteristic estimates by texture and organic matter for hydrologic solutions. Soil Science Society of America Journal, 70(5), 1569-1578.